



Military Institute of Science and Technology

2025-2026 AIAA Design, Build, Fly Proposal



1. Executive Summary

The proposal outlines MIST DBF Team's plan to design, manufacture, and test its banner-towing bush plane, "HeisenBird", for the 2025–2026 AIAA Design, Build, Fly (DBF) Competition. The objective is to develop an aircraft capable of stowing a banner externally during take-off, deploying it remotely in flight, and releasing it after crossing the finish line, while also conducting charter flights to earn revenue. The aircraft is limited to a 5 ft wingspan, 100 W-h battery capacity, and a banner at least 10 inch long with a maximum aspect ratio of 5. The mission requirements call for simple, efficient mechanisms and optimized weights of the aircraft, banner, and deployment system for maximum score and flight stability.

A score-sensitivity analysis identified Mission 2 and Mission 3 lap times as the most significant scoring parameters, followed by banner length and payload mass. Hence, the design prioritized minimizing lap time while maintaining optimal banner length and payload capacity. Based on trade studies, design optimization code results and sensitivity analysis, the team decided on a low-wing monoplane of 60 inch span with a Clark Y airfoil, conventional horizontal and vertical tails with NACA 0009 airfoil, a single motor and propeller, mounted at the nose in a tractor configuration, a semi-monocoque fuselage, a tricycle landing gear, a fuselage aft-belly cavity for efficient banner housing, a removable claw type banner deployment module and release mechanism, and a top-compression balsa plate as the primary passenger restraint system.

The HeisenBird, with a maximum empty weight of 9 lbs, is expected to carry 4 lbs of payload (24 passengers, 8 cargos), completing 10 laps within 5 minutes for M2 and tow a 60 inch polyester mesh banner to complete 8 laps within 5 minutes for M3. To achieve these objectives, the team is organized into two departments with six technical subteams, each with designated leads and personnel, to execute design, fabrication, and testing in an iterative process, delivering an efficient and high-scoring aircraft.

2. Management Summary

2.1 Organization

The 2026 MIST DBF Team consists of 43 undergraduate members, including 24 seniors and 19 underclassmen. The team is fully student-led and is advised by 3 faculty members who provide guidance and share technical expertise. Given the complexity of this year's project, a large team reduces individual workload and allows specialized task assignments. The team is organized into two departments based on functionality: technical and administrative, as shown in Fig. 1. The technical department consists of six technical sub-teams, each with 4-8 members lead by a sub-team lead, while the chief engineer serves as the technical head. The administrative department consists of a treasurer and a media coordinator with the project manager as administrative head. Sub-team leads are responsible for delegating tasks, monitoring progress and arranging respective sub-team meetings.

This team structure establishes a clear chain of command, efficient workflow and hands-on learning opportunities for underclassmen. The entire DBF team meets weekly for collaborating on decisions regarding design, manufacturing, testing and documentation, while each technical sub-team meets twice weekly to monitor progress on individual sub-team member's tasks. The responsibilities and skills of each technical sub-team and all personnel are listed in Table 1.

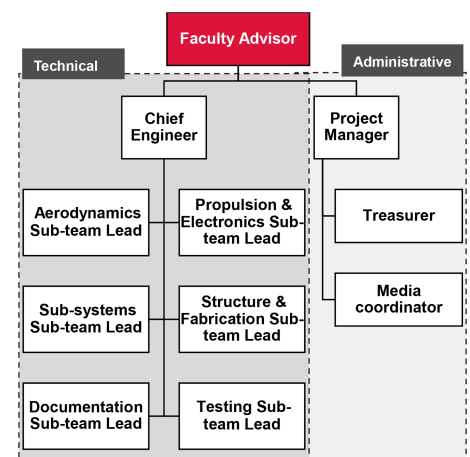


Figure 1. Team organization

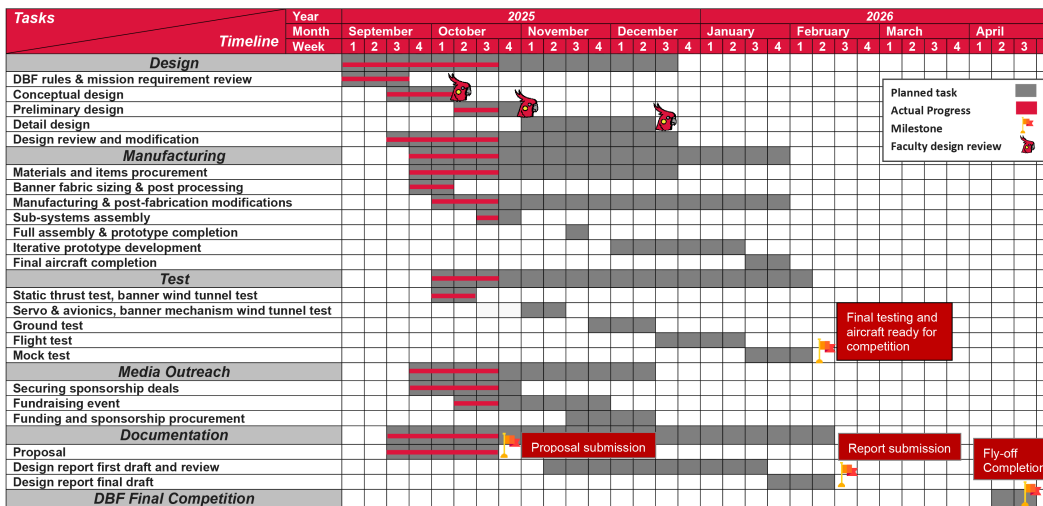
Table 1. Sub-team responsibilities and skills

Sub-teams / Personnel		Responsibilities	Skill Requirements
Technical sub-teams	Aerodynamics	- Perform aerodynamic analysis for sizing, mission performance, stability and control - Conduct trade studies, sensitivity analysis and component optimization - Develop CAD models and run aerodynamic simulations	- Knowledge of aerodynamic principles, stability and control concepts - Proficiency in SolidWorks, MATLAB, XFLR5, ANSYS and CFD tools
	Propulsion and electronics	- Determine optimal motor, battery, propeller and ESC through and performance analysis - Design wiring and integrate propulsion and electronic systems	- Knowledge of propulsion, power and electrical systems - Proficiency in MATLAB, eCalc, MotoCalc, Proteus
	Structure and fabrication	- Design and optimize the aircraft structural framework - Conduct structural analysis, select suitable materials and run structural simulations - Manufacture and assemble all components and sub-systems ensuring safety compliance	- Knowledge of material properties, aircraft structures and different manufacturing methods - Proficiency in SolidWorks, MATLAB, handling machine tools
	Sub-systems	- Design and optimize payload restraint and banner stow/deploy/release mechanisms - Ensure all aircraft sub-systems meet the requirements and follow DBF rules	- Knowledge of servo-based mechanisms and mechanical design - Proficiency in SolidWorks, ANSYS and MATLAB
	Documentation	- Maintain records of important design data and modifications - Prepare and format proposals, design reports following DBF requirements	- Technical writing and formatting skills - Proficiency in LaTeX, MS Word and MS Excel
	Testing	- Test individual components, sub-system mechanisms, propulsion and avionics systems - Record aerodynamic and performance data for further improvement during flight tests	- Test instrumentation setup, data acquisition and analysis skills - Proficiency in MATLAB, Python, MS Excel
Technical personnel	Chief Engineer	- Guide and oversee all engineering design, analysis, fabrication and testing tasks - Ensure coordination between all technical sub-teams and faculty advisors - Coordinate with project manager to align design goals and deadlines - Resolve conflicts between the technical sub-teams	- Leadership planning and analytic skills - Broad technical knowledge in aerodynamics, structures, propulsion and electronics - Proficiency in FEA, CAD and programming tools
Administrative personnel	Project manager	- Oversee workflow, budgeting, media outreach, scheduling and submissions - Arrange team meetings and coordinate between media, treasury and faculty advisors - Align timeline, resource constraints with technical decisions with Chief Engineer - Resolve conflicts within other administrative personnel	- Strong organizational and leadership skills - Experience in quality control and management - Proficiency in MS Word and MS Excel
	Treasurer	- Manage funds, plan budgets, distribute resources maintaining financial compliance - Track all expenses and keep records	- Financial management and organizational skills - Proficiency in MS Excel
	Media coordinator	- Manage sponsorships and funding activities - Coordinate public engagement and publications	- Communication, networking and marketing skills - Proficiency in image and video editing tools

2.2 Milestone Chart

The Gantt chart in Table 2 represents the team's schedule. The project manager references the chart in each full team meeting to ensure timely completion of all the scheduled activities and milestones.

Table 2. Gantt Chart



Key dates: Proposal submission deadline- 31/10/25, Design report submission deadline- 20/02/26, DBF fly-off- 16/04/26 to 19/04/26

Table 3. Budget breakdown

Category	Items (X=Quantity/Type)	Estimated Cost (USD)
Structure	Balsa sheets, carbon fiber rods, fiberglass cloth, adhesives/fasteners	\$2,500
Propulsion	Brushless motors (2), ESCs (2), propellers (6 incl. spares)	\$1,300
Sub-systems	Banner fabric, Servos (15), receivers (3), transmitter (2), LiPo batteries (6), chargers (2)	\$1,200
Fabrication	3D printing filament, CNC/laser runs, consumables (saws, sandpaper, tapes)	\$1000
Testing	Test tools & spares	\$400
Safety	Battery charging sacks, fireproof bags	\$100
Supplies	Workshop hand tools, misc. expendables	\$500
Logistics	Sub-total aircraft	\$7,000
	Travel & accommodation (airfare, hotel, meals, transport), Fly-off registration	\$31,000
Miscellaneous	Delivery and shipping expenses, practice aircraft models, books	\$3,000
	Sub-total operational	\$34,000
	Total	\$41,000
	Contingency (10% for unexpected repairs)	\$4,100
	Grand Total	\$45,100

2.3 Budget

Based on prior experience in DBF competitions, a detailed budget breakdown and estimated costs are presented in Table 3. The primary sources of funding include member contributions and support from the MIST R&D Wing, with an additional \$7,000 expected from fundraising events. The team benefits from access to manufacturing tools in MIST Labs, including 3D printers, CNC, and laser cutting machines, which significantly reduce manufacturing costs, although consumables need to be outsourced. As approximately 15 team members will travel from Bangladesh to Wichita, Kansas, and reside there during the fly-off period, travel and accommodation costs represent the largest portion of the budget, roughly 68% of the total project cost. The next major expenses are delivery and shipping, followed by structural components.

3. Conceptual Design Approach

3.1 Mission Requirements

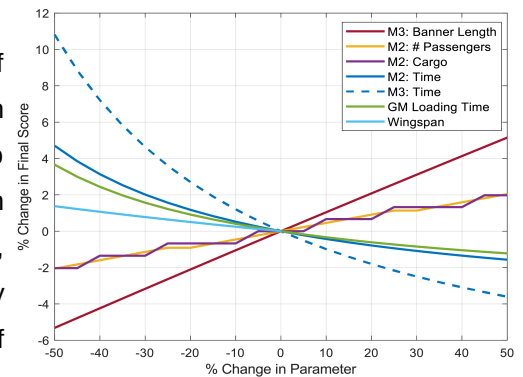
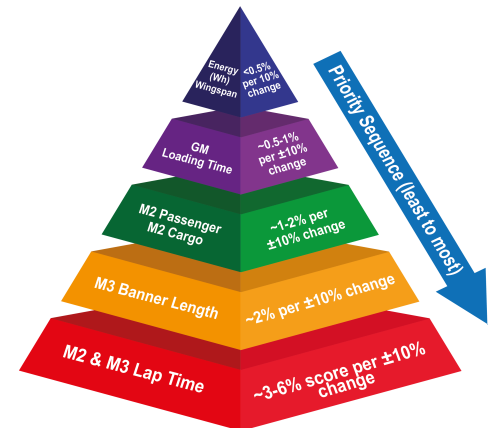
The DBF 2026 competition consists of total four missions: one ground mission and three flight missions. The mission requirements of each mission and their consequent sub-system requirements are summarized in Table 4.

Table 4. Mission and sub-system requirement

Mission	Scoring	Requirement	Sub-system Requirement
Test flight	$M1 = 1.0$	Complete 3 laps within 5 minutes and land successfully with no payload	- Reduce stall speed and empty weight for stable, controllable and energy-efficient flight performance - Ensure batteries provide reliable thrust for safe flight
Charter flight	$M2 = 1 + \left[\frac{N_{Net_Income}}{Max_{Net_Income}} \right]$	Carry declared payload (ducks and pucks) and complete maximum number of laps within 5 minutes, followed by a successful landing	- Optimize number of passenger and cargo for maximum score without compromising stability and speed - Secure passenger and cargo using robust restraint system, along with rigid locks and hatches - Optimize propulsion and battery size to balance thrust and endurance while minimizing the Efficiency Factor penalty
Banner flight	$M3 = 2 + \left[\frac{(N_{laps} \times N_{ten_banner})}{RAC} \right] \left[\frac{(Max_{laps} \times Max_{ten_banner})}{RAC} \right]$	- Banner must be at least 10 inches long with a maximum aspect ratio of 5 - Stow banner at takeoff, remotely deploy after first upwind turn maintaining vertical orientation in flight, release after final lap, and complete this mission within 5 minutes followed by a successful landing	- Design a robust, servo-driven banner deployment and release system - Selection of banner configuration and fabric for low drag and maximum score - Ensure propulsion system produces high thrust to overcome banner induced drag - Optimization of wingspan to balance aerodynamic performance and scoring, while minimizing RAC penalty and maintaining 5 ft span limit
Ground mission	$GM = \frac{Min_{mission_time}}{N_{mission_time}}$	- Timed installation and removal of declared payload (ducks and pucks) - Timed installation of the banner in stowed configuration, with ground demonstration of its deployment and release	- Build easily accessible and simple passenger and cargo restraint systems for quick loading and secure operation - Construct a quick banner attachment, simple and fast banner deployment and release mechanism.

3.2 Sensitivity Analysis

Initial propulsion studies, banner wind tunnel tests and general understanding of aircraft performance were used to establish correlations with 2026 DBF mission scoring parameters, which were then applied to assess total score sensitivity to variations in each parameter using a custom MATLAB code. Results shown in Fig. 2, reveal that lap time is the most impactful parameter for both M2 and M3, as small reductions in lap time yield substantial score gains. It is followed by ground mission time, which translates directly into overall score in the form of faster loading and unloading process. It is observed that, increasing payload mass in the form of additional passengers or cargo raises the score initially due to higher net revenue, but eventually slows the aircraft and raises lap time producing diminishing returns. Again, making the banner longer improves the score but adds drag and mass which negatively impact lap performance. The optimal banner length is near the 60 inch baseline, beyond which gains are minimal. Moreover, Rated Airplane Cost (RAC) is found to have minimal influence on overall score, making wingspan relatively insensitive; however, shortening the span would increase induced drag and wing loading, reducing lift efficiency, endurance and overall mission performance. Therefore, further efficiency gains are best pursued through wingtip shape, twist and weight optimization, rather than decreasing span.

**Figure 2. Mission sensitivity results****Figure 3. Sensitivity pyramid**

The ranked pyramid in Fig. 3 provides an explicit visualization of how each of the parameters affects DBF 2026 scoring, establishing relative importance of each design parameter, indicating how they must be prioritized in design. Overall, total score maximization depends primarily on lap time minimization through aerodynamic and propulsive efficiency, optimizing turns and mission execution strategy and reducing GM time through efficient loading operations. This balance between payload and performance has led to the selection of 24 rubber ducks and 8 hockey pucks as the baseline configuration for subsequent design phases.

3.3 Trade Study

Prior to design optimization, trade studies were conducted to establish the initial configuration of the aircraft. Each design option was evaluated using a 1-10 rating scale against key performance metrics for every component, enabling systematic comparison of alternatives. Trade studies covering banner deployment mechanism, banner storage location, wing location and passenger restrain system are shown in Table 5, as they dictate the overall design of the aircraft significantly. These configurations will be further justified in the next section.

Table 5. Trade study of design alternatives

Banner Deployment Mechanism					Banner Storage Location				
Parameters	Weight (%)	Claw Mechanism	Rubber Band Release	Pivot-Rod Trigger Mechanism	Parameters	Weight (%)	Aft-belly cavity	Underwing Mount	Tail-Mounted Pod
Reliability	35	9	5	8	Aerodynamic Drag	30	9	7	6
Actuation Speed	20	8	9	9	Deployment Reliability	25	9	8	7
Structural Weight	15	7	9	6	Ground Clearance	20	8	7	9
Operational Repeatability	20	9	6	8	Structural Simplicity	15	7	8	8
Manufacturability	10	8	9	8	Manufacturability	10	8	9	8
Total Weighted Score	100	8.4	7.0	7.9	Total Weighted Score	100	8.4	7.6	7.4
Wing Configuration					Passenger Restraint System				
Parameters	Weight (%)	High-Wing	Mid-Wing	Low-Wing	Parameters	Weight (%)	Top Compression Plate Restraint	Velcro Straps	Foamed Cutouts
Aerodynamic Stability	25	10	8	7	Restraint Efficiency	35	9	6	8
Payload Accessibility	25	3	5	8	Ease of Loading/Unloading	20	9	7	6
Flight Speed	20	6	7	8	Weight	15	7	9	6
Payload Capability	20	9	7	9	Structural Integrity	20	9	6	8
Assembly and Maintenance	10	8	5	8	Manufacturability	10	8	9	5
Total Weighted Score	100	7.05	6.55	7.95	Total Weighted Score	100	8.6	7.5	6.9

3.4 Preliminary Design

Based on score sensitivity analysis and trade study, preliminary design was conducted to establish component specifications and dimensions (outlined in Table 6) for a light-weight aircraft capable of towing a large banner. All aircraft parameters were derived from a MATLAB-based design optimization code developed using payload requirements and banner dimensions as input. While typical bush plane, having high-wing and tail-dragger configuration, prioritizes rough field operations and ground clearance, this design focuses on aerodynamic efficiency, endurance and fast payload handling during GM. To meet these objectives, a low-wing configuration was selected to allow top-loading access to the passenger compartment and provide a deeper fuselage cross-section for increased payload capacity without widening the airframe. Given the low-wing setup, a nose-mounted tractor configuration combined with a tricycle landing gear, was chosen to ensure adequate ground clearance and taxi control. The wing utilizes the maximum allowable 5-ft span to allow a larger aspect ratio, thus reducing induced drag from wingtip vortices and improve lift efficiency. A rectangular inboard wing section was integrated into the fuselage for ease of construction, while a tapered outboard section further minimizes induced drag. Winglets were added to reduce induced drag, improve roll and yaw stability and minimize vortex interference with tail and banner. A dorsal fin was added to improve yaw stability and reduce sideslip tendencies during banner tow. A 3° dihedral angle is employed for better lateral stability and handling. XFLR5 analysis showed that, for level flight, 0° angle of attack provides the lowest induced drag and maximum practical cruise speeds of 108 ft/s in M2 and 89 ft/s in M3. Anticipating the significant induced drag of a large banner, the propulsion system was chosen to deliver both high thrust and endurance to sustain the M2 and M3 duration and counter the increased power demand in gusty conditions. A high efficiency, high torque and lightweight motor paired with a large-diameter, large-pitch propeller was necessary, to maximize both thrust and cruise speed, justifying the selection of a BLDC motor: Scorpion A-4220-540kv V2. Efficiency Factor (EF) showed minimal sensitivity, so a 6S 4500 mAh LiPo was selected for longer endurance despite minor weight addition. To further enhance aerodynamic efficiency, the fuselage was shaped to minimize frontal area and maintain smooth contours, while preserving internal capacity. A semi-monocoque fuselage arrangement was employed to maximize internal capacity, thus increasing payload accommodation and M2 score. The square-rounded cross-section, with rearward up-sweep and side-sweep, balances structural efficiency with low aerodynamic drag, resulting in reduced power consumption during flight.

3.4.1 Airfoil and Banner Selection

A trade study of 17 airfoils, including high-performance candidates like the SD7032 and MH84, was conducted, considering factors like lift-to-drag ratio, stall characteristics and pitching moment and Clark Y was selected for its superior balance of performance and practicality. Its 11.7% thickness-to-chord ratio allows sufficient space for spar integration and still provides mild stall behavior to ensure safety during M3 at low

Table 6. Preliminary design sizing results

Wing		Tail		Mission Performance	
Span	60.00 in	Horizontal tail span	24.49 in	M1 (time)	95.00 s
Mean Aerodynamic Chord	13.49 in	Horizontal tail MAC	9.80 in	M2 (no. of laps)	10
Aspect ratio	4.50	Vertical tail span	7.88 in	M3 (no. of laps)	8
Area	800.00 in ²	Vertical tail MAC	7.92 in	Cruise Speed (M2)	108 ft/s
Airfoil	Clark Y	Airfoil	NACA 0009	Cruise Speed (M3)	89 ft/s
Propulsion system		Aircraft weights		Banner	
Propeller	16X8	MTOW (M1 & M3)	9 lbs	Length	60.00 in
Battery	4500 mAh	Payload	4 lbs	Width	12.00 in
Max thrust	10.57 lbf	Total weight (M2)	13 lbs	Weight	4.23 oz

speeds. NACA 0009 symmetrical airfoil was selected for both the horizontal and vertical tail due to its linear control response and suitability for safe banner-tow maneuvers. Banner size and material was optimized through wind tunnel testing across various fabrics, such as LDPE, Kevlar, Tyvek and Polyester Mesh Fabric (PMF), both with and without fringes at various speeds, to quantify induced drag and flutter behavior. The tests showed a drag coefficient range of 0.13–0.18, where fringes reduced flutter but increased drag. PMF was selected as it demonstrated minimal flutter without fringes and the lowest drag. A MATLAB-based optimization code indicated that a 60 inch banner with aspect ratio of 5 could be towed at 82-98 ft/s, completing approximately 8 laps within a 5-minute flight window.

3.4.2 Banner Stowing, Deployment and Release Mechanism

The banner deployment system, shown in Fig. 4, consists of a removable claw-type module mounted in a fuselage cavity near the aircraft's center of gravity (CG), a servo-controlled locking pin and a high-strength fishing line used as the tow rope. The aft-belly cavity houses the rolled-up banner within its recess maintaining aerodynamic shape as well as qualifying as external stowage system. This module is used only in M3 to simplify configuration and minimize drag in other missions.

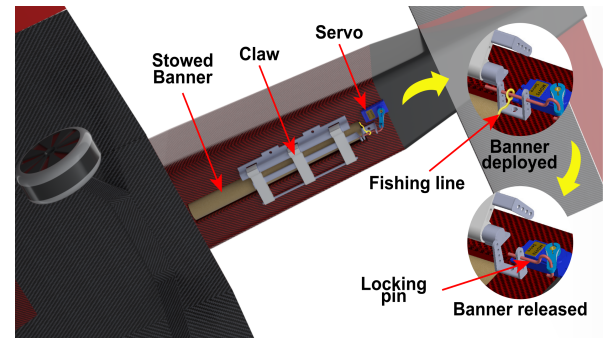


Figure 4. Banner mechanism

The servo operates in three positions to control banner phases: full extension, where the pin locks into the claw, securing the banner on take-off; mid-position, where the pin partially retracts, allowing the claw to open and drop the banner under gravity, while the airflow assists to unfurl the banner; and full retraction, where the pin releases the tow rope end, freeing the banner completely from the aircraft. A thin, rigid stick is fixed along the banner's leading edge, with a weighted capsule containing eight M6 nuts at one end to maintain vertical orientation during flight. Positioning the stowage cavity and tow-point near the CG minimizes post-deployment changes in longitudinal trim and avoids destabilizing moments.

3.4.3 Payload Accommodation and Restraint Systems

The fixed cargo compartment, as shown in Fig. 5, is built into the fuselage just above of the wing box and is accessed via a hinged cargo door rigidly attached to the fuselage. 8 standard hockey-pucks are carried side-by-side in a prone orientation in an easily removable holding cargo box with a sliding closure panel along its width to prevent side-to-side movement of pucks. When inserted, the cargo door is sealed and locked by a simple mechanical latch system. Directly over cargo bay, the passenger compartment is completely isolated from other interior sections, as shown in Fig. 6 and 7, containing 24 rubber ducks in a 12×2 configuration. Entry and exit are facilitated through a fuselage-hinged door opening only into passenger bay. The seating base is a rectangular balsa plate with machined grooves matching duck dimensions, secured to the floor with screws. Removable sticky pads at the bottom of each groove ensure a secondary restraint. The primary restraint system is a second balsa plate with matching rectangular cutaways through which the duck heads protrude, slightly compressing their bodies against the seat plate to fit snugly. The restrain plate is secured at its front and back using quick-release clip locks that secure the plates together and permit all passengers to be restrained or released simultaneously during ground operations.

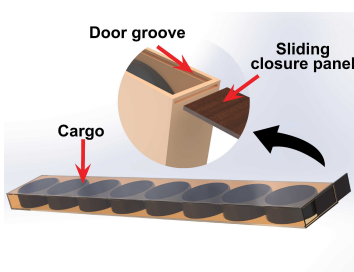


Figure 5. Cargo box

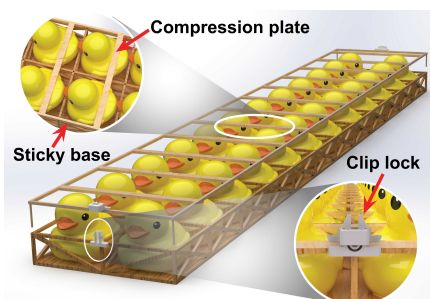


Figure 6. Passenger restraint system

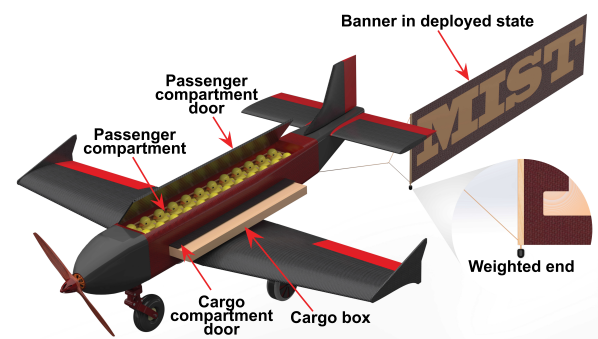


Figure 7. Render of the proposed aircraft

4. Manufacturing Plan

The intended manufacturing plan, shown in Fig. 8, highlights how manufacturing, testing and design modifications will proceed in parallel with detail design. Before fabrication, finite element analysis and other computational tools will be used to identify structural and design issues, minimizing manufacturing errors and cost. The wing and tail will be constructed using laser-cut balsa ribs with carbon fiber spars, covered with MonoKote film for a smooth, lightweight finish. The fuselage will have a semi-monocoque structure with bulkheads, longerons and stringers, covered with MonoKote and locally reinforced with carbon fiber at landing gear, motor and banner mounting points. The passenger compartment will be made of lightweight balsa panels. The banner claw unit will be 3D printed using carbon fiber-reinforced PLA for strength and low weight. Iterative prototyping and testing will guide design improvements, producing successive prototypes that will eventually evolve into the final aircraft.

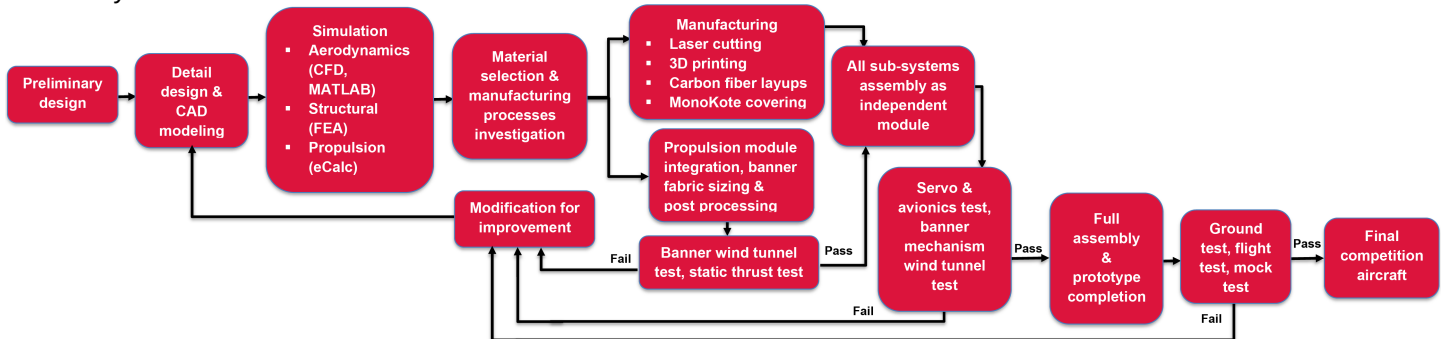


Figure 8. Manufacturing plan flowchart

5. Testing Plan

As outlined in Table 7, testing will proceed in stages, starting with individual systems and modules. Only after their successful verification, the fully assembled prototypes will be tested. Ground test will first assess the structural integrity of aircraft and validate the performance of payload restraint and banner mechanisms and any weaknesses will be addressed. Upon successful ground testing, flight tests will be conducted to assess overall aircraft performance and identify potential areas of improvement. Finally, mock tests will replicate the main competition missions and critical data from sensors and visual observations will be utilized to refine performance and finalize the aircraft for the competition.

Table 7. Testing plan chart

Test Type		Goal	Test Method
Propulsion & electronics test	Static thrust test	Determine most efficient motor, propeller, ESC and battery combination and validate theoretical predictions	Measure thrust and current draw for different propeller sizes with various battery and motor combinations using a motor test stand and compare results with eCalc/MotoCalc data
	Servo and avionics test	Verify control response and reliable TX-RX communication	Connect the motor and receiver to the battery, and observe propeller and control surface responses while moving the transmitter away, up to a minimum distance of 2000 ft.
Wind tunnel test	Banner test	Evaluate banner material and sizing for minimum drag, weight and fraying	Mount banner samples in wind tunnel, vary the wind speed (30-110 ft/s) and record drag forces using load cell
	Banner mechanism test	Validate reliability of stowage, deploy and release mechanisms	Mount mechanism mockup in wind tunnel, simulate take-off and cruise speeds, actuate servo systems using transmitter and confirm smooth deployment and detachment
Ground Test	Banner test	Reconfirm banner drag and durability under dynamic airflow	Mount banners on car roof, vary the car speed (50-130 ft/s) and record drag forces using load cell
	Restraint test	Ensure reliable attachment of cargo and passenger	Load ducks and pucks, simulate in-flight vibration by shaking and observe for displacement or loosening
	CG test	Verify CG is in the intended location and aircraft is stable	Lift the aircraft by wingtips, observe stability, make adjustments until aircraft is stable and CG is found
	Load test	Verify structural integrity of wing and fuselage under large centered loading	Install the aircraft with maximum payload, mount it on test stand and apply extra weight at its CG
	Landing impact test	Ensure landing gears can withstand impact loads safely	Drop the aircraft from about 3 ft and inspect for structural damage
	Assembly & Taxi test	Ensure efficient, robust assembly and controlled taxiing	Assemble the whole aircraft and simulate in-flight vibration and check for damage or loose joints, perform straight taxi runs on ground and verify steerable landing gear response
Flight test	Cruise speed test	Verify cruise speed with and without payload	Fly the prototype and record its cruise speed using the telemetry system
	Steep turn test	Verify minimum safe bank angle and turn radius	Perform steep turns with minimum bank angle of 20°, verify stable response and record data using telemetry system
	Banner test	Compare in-flight banner drag with ground test results and finalize banner configuration	Measure drag forces using strain gauge mounted on the test prototype while in flight
	Stability test	Confirm stable flight in all flight conditions	Fly the prototype, perform various maneuvers and record pilot's feedback and visual cues
	Full system test	Verify all systems and mechanisms function as intended	Perform full mission attempts with the test prototype
Mock test	Ground mission test	Verify efficient loading/unloading and banner operations on ground	Perform several ground mission cycles under timing constraints and check for any complication or mechanism fatigue effects
	M1 flight test	Validate unloaded flight performance	Perform 3 laps within 5 minutes, land successfully and checking for any complication
	M2 flight test	Validate loaded flight performance and endurance	Fly with declared number of payloads for 5 mins, record current draw, lap number and lap time
	M3 flight test	Validate in-flight banner stow-to-release sequence, stability and alignment under aerodynamic loads	Perform full M3 sequence within 5 mins, monitor drag, stability, banner alignment and telemetry data